



Talk Dirt to Me: A Tale of Three Compost Sources

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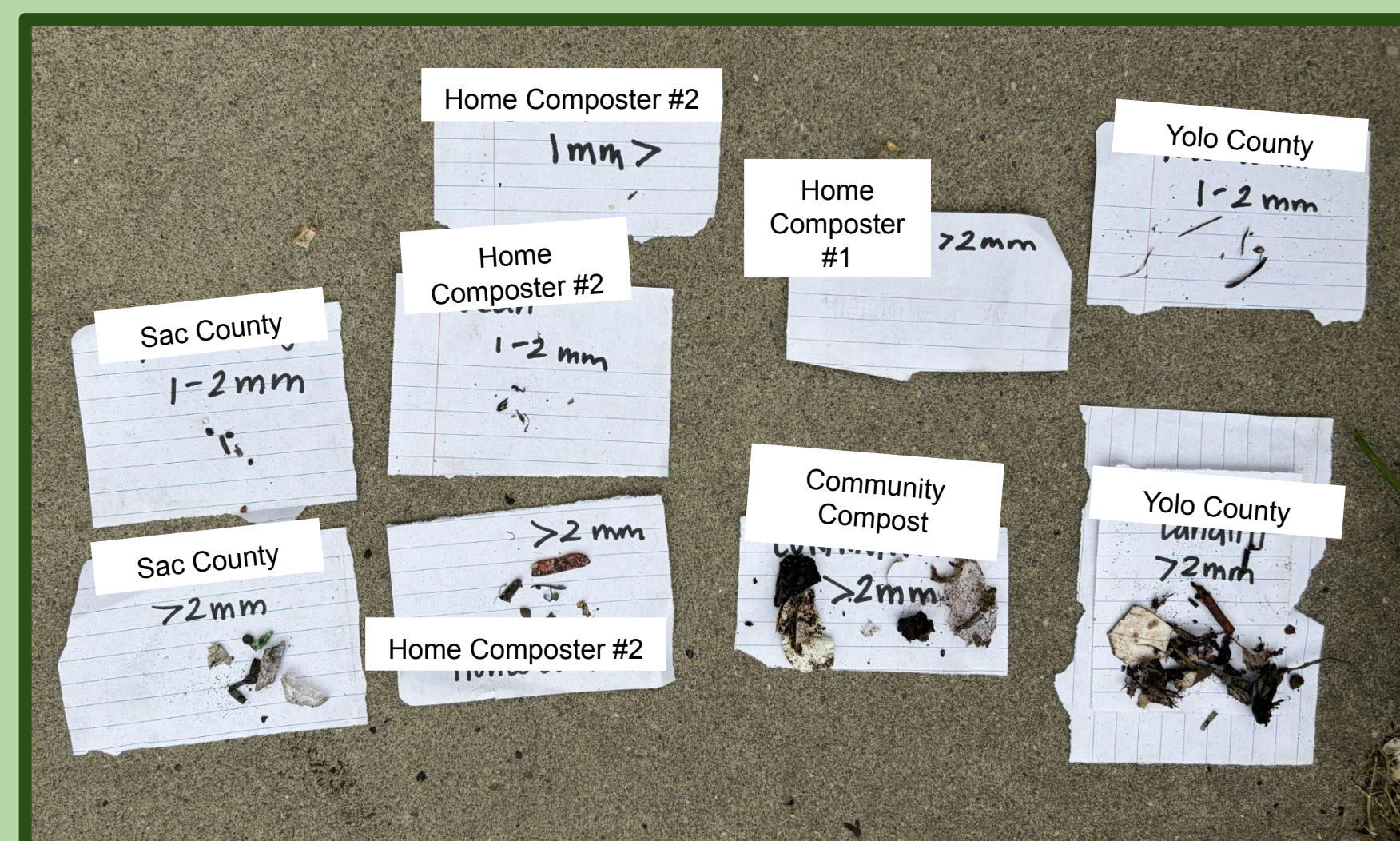


Introduction

Research on composting is very important to today's world through a number of aspects. For example, it is important for communities because it can help them with their own projects--whether at home or in their community. Having too much nitrate or too many plastics can make a vital difference in their plants, so it is effective and valuable knowledge. Not only that, it can also help us as we battle climate change. Composting is one of many sustainable cycles that exist. This means that the scraps of the food we eat are use to make compost, the compost helps grow plants, and the scraps of the plant that grow help continue to cycle. Composting is important!

Hypothesis

1. If a compost is sourced from a large facility, then the amount of plastics found in the compost is expected to increase.
2. If a compost is sourced from a large facility, then the amount of nitrogen (nitrate) is expected to increase.



All compost samples' plastic data

Materials & Methods

- We collected five samples of compost from three size scales: home compost (small), community compost (medium), and municipal compost (large).
- Both the home and community compost samples were managed manually, while the municipal sources were produced mechanically. All sources were produced in either Yolo County or Sacramento County.
- To measure nitrate concentration, we used nitrate testing strips. We mixed the compost in a 0.01M calcium chloride dihydrate solution by shaking, let it sit, and then sunk a piece of the testing strip in the water. After a minute, we were able to determine the nitrate levels by the color of the piece dipped in the water.
- To measure the microplastics in the composts, we used 500 grams of each compost, and sifted each through a 1mm and 2mm sieve to categorize the plastics by size. We then calculated the plastic concentration by taking the weight of the plastics and dividing it by 500g.

Results

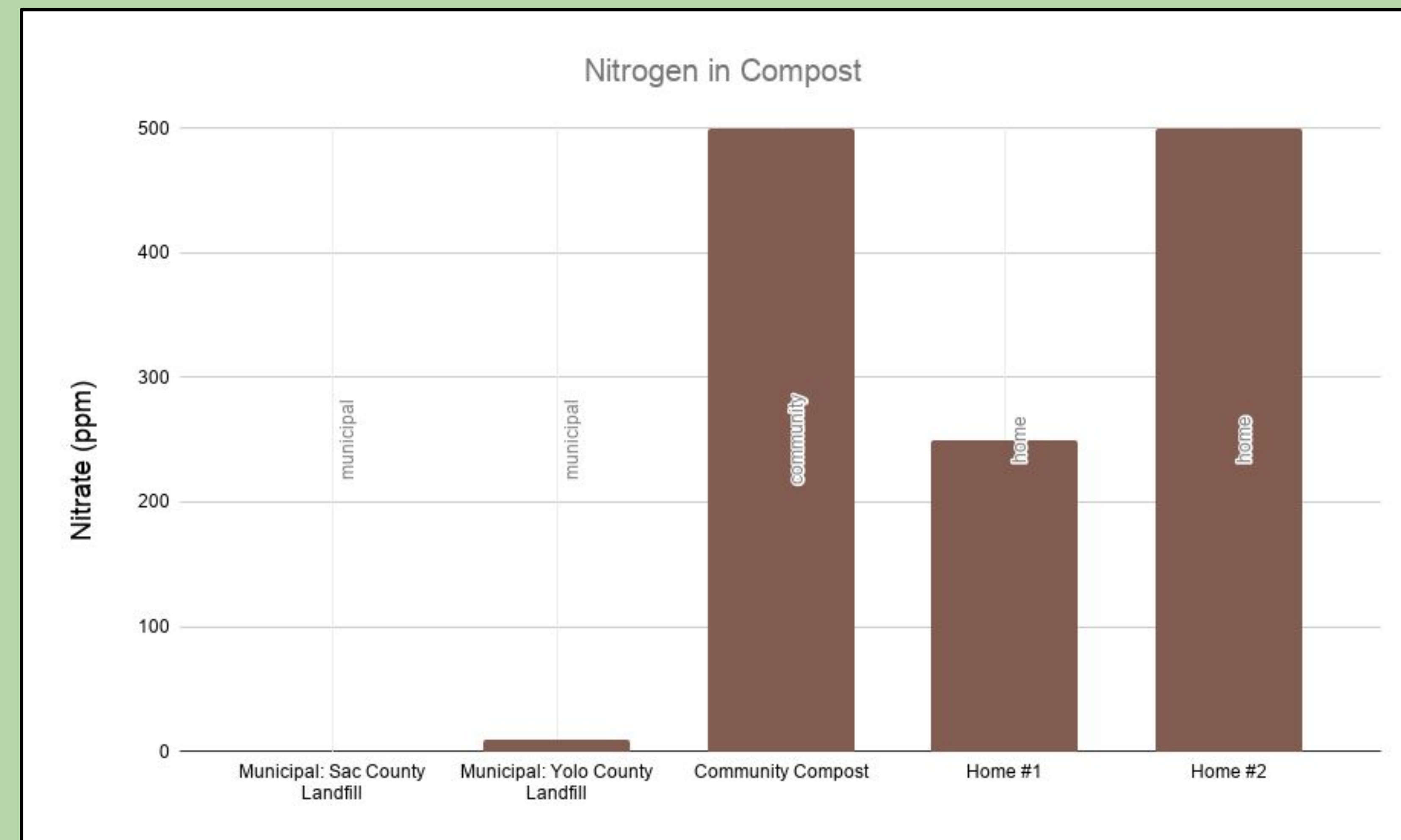


Figure 1: Compost Nitrogen Content. Home and community composts have higher nitrogen than landfills

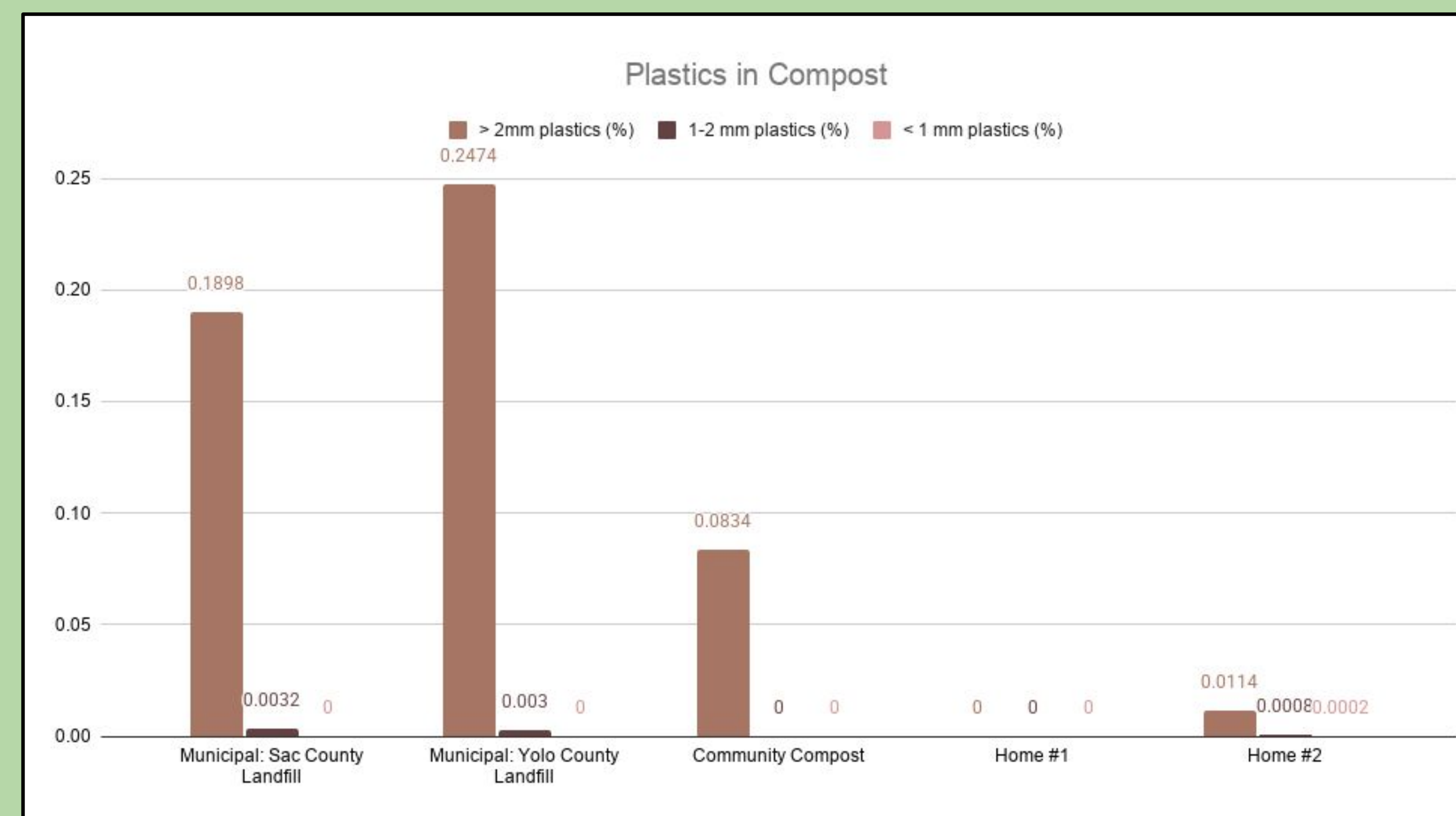


Figure 2: Compost Plastic Content. Landfills have more plastic compared to community and home composts

Discussion

There may be a difference in the compost type because of the different scales each composting operation. For example, home compost is mostly reliant on the person whose making the compost. However, community compost uses greens and browns from all of the community. This could also be a way for stickers on peels to get in the compost. Additionally, a landfill would have the most amount of resources. This means that their compost would be expected to be the most balanced. However, through our lab results, this is clearly not the case.

There may be less nitrate in the landfill compost because of the quality of the compost materials. It takes a lot longer for someone to gather their food scraps and drop them off at a landfill than it does for someone to put their food scraps in their compost in their backyard. Therefore, it makes sense why the home and community composts have higher nitrogen, and the landfills have lower nitrogen percentages.

For trends, it is clear that the home and community composts have higher nitrogen and lower plastics when compared to the landfills.

I would recommend community members to make their own compost or get it from a community source because it provides nitrogen nutrients and improves soil structure. Home or community composts are generally seen as better in this scenario because of their contamination control, nutrient value, and long term health benefits for soil.



Sifting compost & weighing plastic.

References

- CalRecycle (n.d.). Home Composting. Accessed May 3, 2026. <https://calrecycle.ca.gov/organics/homecompost/>
- EPA (2025). *Composting at Home*. Reduce, Reuse, Recycle. Accessed May 3, 2026. <https://www.epa.gov/recycle/composting-home>.
- Soil Science Society of America (n.d.). *Soil and the Environment*. Soils 4 Teachers. Accessed May 3, 2026. <https://www.soils4teachers.org/soil-and-environment/>
- Soil Science Society of America (n.d.). *Soil Basics*. Soils 4 Teachers. Accessed May 3, 2026. <https://www.soils4teachers.org/soil-basics>
- Soil Science Society of America (n.d.). *Soil Fertility*. Soils 4 Teachers. Accessed May 3, 2026. <https://www.soils4teachers.org/fertility/>

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